SHAIK IBRAHIM

CYBERSECURITY ANALYST

SECURITY ENGINEERING & AUTOMATION

Defensive network analysis / SOC workflows / DevSecOps
Security software development / Self-hosted systems

SELECTED ENGINEERING WORK

nScout / **SentinelLab** / **CERBERUS**

Locat / CaseTrace / SENTINEL EDGE

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01 / PROFESSIONAL PROFILE

Security-focused engineering with an investigative mindset.

B.Tech in Computer Science and Engineering from GITAM University, with Cybersecurity and Ethical Hacking training from Boston Institute of Analytics. Focused on defensive security, network traffic analysis, security operations, vulnerability assessment, automation and software engineering.

Independent engineering work includes network packet investigation, experimental SOC/SIEM tooling, local-first security scanning, self-hosted encrypted communications and early-stage digital investigation systems. Across these projects, I work with application architecture, Python services, React interfaces, security controls, testing, deployment, technical documentation and release workflows.

Interested in cybersecurity analysis, security engineering, threat detection, DevSecOps and security-focused software development. Project maturity, validation status and future work are identified separately throughout this document.

02 / TECHNICAL SKILLS

NETWORK & SECURITY

PCAP analysis / TCP/IP / DNS, HTTP & TLS / threat detection / investigation workflows

SECURITY TOOLS

Wireshark / Nmap / Burp Suite / Metasploit / OWASP ZAP / CyberChef

APPLICATION ENGINEERING

Python / JavaScript & TypeScript / React / FastAPI / REST APIs / WebSockets

PLATFORMS & DELIVERY

Linux / Windows / Docker / GitHub Actions / Tauri / Raspberry Pi

DATA & SELF-HOSTING

MongoDB / MariaDB / IndexedDB / SSH / SMB & NFS

SECURITY WORKFLOWS

Detection rules / vulnerability assessment / reporting / evidence handling

FEATURED ENGINEERING PROJECT

nScout

NETWORK INTELLIGENCE & PACKET INVESTIGATION

Python / FastAPI / Scapy / React / WebSockets / MongoDB / PyInstaller

PROJECT OVERVIEW

nScout is a defensive network-monitoring and investigation application that converts packet captures into host, connection, protocol and security context. It supports live capture, imported PCAP files and simulated traffic, helping an analyst move from network summaries to the packets underlying an observation.

PACKET AND CONNECTION ANALYSIS

- Inspect Ethernet, ARP, IPv4/IPv6, TCP, UDP, ICMP, DNS, visible HTTP and observable TLS metadata, including protocol fields, TCP flags, timing and hexadecimal/ASCII views.
- Reconstruct bidirectional conversations; correlate DNS answers, TCP lifecycle events and visible TLS/HTTP activity; examine retransmissions, duplicate ACKs, resets and other TCP health indicators.

DETECTION AND INVESTIGATION

- Generate heuristic investigation leads for scan-like behavior, failed connections, ARP identity changes, unusual DNS activity, traffic spikes and beacon-like timing.
- Provide finding severity, confidence and evidence; build observed host profiles, bounded risk scores, correlated timelines and Boolean packet searches.
- Export investigation results as JSON, standalone HTML and PDF; offer optional AI explanations without making core analysis dependent on AI.

ENGINEERING AND VALIDATION

- FastAPI and Scapy backend with React/WebSocket analyst interface; Windows standalone packaging using PyInstaller and Inno Setup.
- Repository documents a v0.2 stable release and v0.3 security-intelligence milestone with 52 focused regression tests. Findings remain analyst leads, not proof of compromise; HTTPS payloads are not decrypted.

DEVELOPMENT ROADMAP / NEXT MILESTONES

- Improve native Windows desktop startup and backend lifecycle management, while preserving an optional development/web mode.
- Research Windows Filtering Platform capture as a possible alternative to mandatory Npcap, subject to feature-parity testing.
- Expand regression PCAP datasets, detection tuning, performance profiling and cross-platform distribution.

PROJECT REPOSITORY

https://github.com/ibrahim1101/nScout

FEATURED ENGINEERING PROJECT

SentinelLab

EXPERIMENTAL SECURITY OPERATIONS PLATFORM

React / Python / FastAPI / MongoDB / Tauri 2 / Docker / GitHub Actions

PROJECT OVERVIEW

SentinelLab is an experimental SOC/SIEM desktop platform designed to combine event collection, normalized security data, detection rules, alert triage, threat hunting and investigation documentation in a single analyst workspace.

SECURITY OPERATIONS WORKFLOWS

- Ingest JSON, JSONL, CSV, Syslog and CEF events; normalize, search, filter, export and inspect event details.
- Manage detection rules, alerts, case notes, investigation tasks, evidence and reports; explore saved hunts and MITRE ATT&CK; coverage.
- Support IOC imports, investigation graphs, detection replay, pipeline observability and approval-gated response playbook workflows.

PLATFORM AND SECURITY ENGINEERING

- Use role-aware authentication, organization/workspace boundaries, audit trails and account-security controls.
- Configure optional local or OpenAI-compatible AI assistance and outgoing webhooks; Discord delivery has been manually tested.
- Package desktop builds through CI for Windows, Linux and macOS, with MongoDB installed separately for desktop use.

MATURITY AND VALIDATION

- Experimental alpha rather than a production-certified SIEM. Windows installation and selected workflows have manual verification; Linux/macOS packaging has CI evidence but not full runtime acceptance.
- Some external intelligence integrations remain placeholders; complete SOC end-to-end testing is ongoing.

DEVELOPMENT ROADMAP / NEXT MILESTONES

- Carry out controlled SOC investigations with reproducible event evidence and professional reports.
- Improve resizable event detail fields and investigation popups; refine exports and analyst usability.
- Expand verified integrations, detection regression coverage and desktop acceptance testing.

PROJECT REPOSITORY

https://github.com/ibrahim1101/SentinelLab

FEATURED ENGINEERING PROJECT

CERBERUS

LOCAL-FIRST DEVSECOPS SECURITY SCANNER

React / Python / FastAPI / Tauri 2 / Local scanning engine / GitHub Actions

PROJECT OVERVIEW

CERBERUS is a desktop-oriented DevSecOps security scanning application built around local project analysis, findings review, dependency intelligence and report workflows. Its design aims to keep scanning and sensitive source material under local control by default.

SCANNING AND INVESTIGATION

- Run a bundled local scanning engine; view findings, dependency information, scan progress, history and project-level results.
- Support cooperative scan cancellation, scan accounting, masked sensitive evidence and JSON/HTML/SARIF-related reporting paths.
- Offer optional OSV dependency-vulnerability lookups, with network use separated from local scanning.

ARCHITECTURE AND SAFEGUARDS

- Integrate React workspaces, a Python FastAPI bridge and a Tauri 2 native shell with packaged Python sidecar.
- Use a loopback-focused bridge and per-launch desktop authentication token; apply path/preview checks and secure credential-storage considerations.

ENGINEERING VERIFICATION

- Automated frontend, backend, bridge and desktop build checks have passed at documented milestones.
- Physical installer acceptance, actual export flows and packaged key-storage behavior require further validation; Linux/macOS preview packaging is experimental.

DEVELOPMENT ROADMAP / NEXT MILESTONES

- Complete real-world Windows installation, scanning, cancellation, export and uninstall acceptance tests.
- Implement opt-in AI finding explanations and secure outbound scan webhooks with privacy previews and protected credentials.
- Improve report usability, dependency intelligence and cross-platform runtime validation.

PROJECT REPOSITORY

https://github.com/ibrahim1101/Cerberus

FEATURED ENGINEERING PROJECT

Locat

SELF-HOSTED ENCRYPTED COMMUNICATIONS

TypeScript / React / Node.js / MariaDB / IndexedDB / Web Crypto / Capacitor

PROJECT OVERVIEW

Locat is a privacy-focused self-hosted messaging project designed for personal infrastructure, including Raspberry Pi and Linux servers. It explores device-side encryption, messaging reliability, account administration and multi-device application delivery.

MESSAGING AND PRIVACY

- AES-GCM, ECDH/HKDF, pinned contact keys, local message history, encrypted backup/restore and retry delivery.
- Group membership controls, key rotation, session revocation, Web Push, administrative controls and Raspberry Pi deployment.
- The repository describes device-side encryption, direct/group conversations and an Android application based on Capacitor. Locat is not a Signal Protocol implementation and does not claim Signal-style forward secrecy.

SELF-HOSTED ENGINEERING

- Work across web/mobile clients, server APIs, local persistence, account handling and deployment on user-controlled infrastructure.
- Locat 1.0 is under active development, including unified mobile and self-hosted application workflows.

EXPANDED PLATFORM DIRECTION

- Develop an integrated local media experience for permitted videos, photos and music, with sorting and browsing across authorized network-connected devices.
- Improve native Android navigation, media sharing from conversations and local-network operation without a required external VPN.

DEVELOPMENT ROADMAP / NEXT MILESTONES

- Validate the unified application and actual encryption/media workflows on Windows, Android and Raspberry Pi.
- Harden multi-device delivery, permissions, secure file access and server deployment.
- Complete media-library discovery, playback, cross-device sharing and stable release testing.

PROJECT REPOSITORY

https://github.com/ibrahim1101/Locat

04 / EARLY-STAGE PROJECTS

CaseTrace

SECURITY INVESTIGATION & CASE MANAGEMENT

Early-stage implementation / React / FastAPI / Python

CaseTrace is an emerging investigation workspace intended to organize cases, evidence, findings, entities, timelines, tasks and reports. The current repository includes frontend code and modular backend routers for authentication, cases, evidence, findings, search, timeline, AI, automation, reporting and administration. These modules establish a development foundation, not proof of complete functional workflows.

DEVELOPMENT PRIORITIES

- Stabilize authentication, database initialization and end-to-end user flows.
- Verify evidence workflows, case relationships, timeline visualization and report generation.
- Introduce secure, auditable AI and automation integrations only after core workflow validation.

PROJECT REPOSITORY

https://github.com/ibrahim1101/CaseTrace

SENTINEL EDGE

EDGE AI SECURITY RESEARCH PLATFORM

Concept and prototype planning / Raspberry Pi 5 8 GB / AI HAT+ 2 (40 TOPS)

SENTINEL EDGE is a planned hardware-assisted security research platform investigating whether security telemetry processing and selected AI inference workloads can operate locally on an embedded Linux system. It is not presented as a completed device or deployed monitoring product.

PROTOTYPE PLAN

- Prepare Raspberry Pi OS, accelerator tooling and repeatable installation procedures.
- Benchmark supported inference workloads and select a narrow, measurable security-monitoring proof of concept.
- Evaluate latency, CPU/RAM usage, power and detection quality before expanding the scope.

Status: planning and architecture; no repository link supplied.

05 / EDUCATION & TRAINING

B.Tech — Computer Science and Engineering

GITAM University, Visakhapatnam | June 2018 – April 2022

Cybersecurity and Ethical Hacking

Boston Institute of Analytics | June 2024 – January 2025

Practical training in vulnerability assessment, password-hash auditing, traffic analysis, controlled attack simulations and Metasploit in isolated Windows/Linux labs.
Android capstone: ADB, bootloader unlocking and operating-system flashing.

06 / CERTIFICATIONS & LEARNING

Cybersecurity and Ethical Hacking

Boston Institute of Analytics — training completed January 2025

Python Programming

University of Michigan / Coursera — 2022

Python Programming

Udemy — 2023

Ethical Hacking

Udemy — 2023

Android Development

GITAM University — 2022

CompTIA Security+

Course preparation only; certification not claimed

07 / ACADEMIC & EARLIER PROJECTS

Facial Recognition Attendance Prototype

Academic Python/OpenCV prototype for student attendance workflows.

Voxel Game Prototype

Python/Ursina scripting experiment inspired by voxel-style game systems.

08 / ADDITIONAL TECHNICAL ACTIVITIES

HOME LAB & SELF-HOSTING

Raspberry Pi 5 Linux server; static LAN addressing, SSH, OpenMediaVault NAS, SMB/NFS shares and Plex. Docker Compose/Portainer services, remote-access experimentation with Twingate, and locally hosted Ollama/Open WebUI. Raspberry Pi/Hailo object-recognition experiments; Frigate/RTSP CCTV integration remains in development.

GAME MODDING & RUNTIME EXPERIMENTATION

Exploration of REFramework and UE4SS modding workflows, game-state customization, user-interface scripting, quality-of-life tools, save-file compatibility and debugging. Experimental concepts and prototypes are not represented as published finished mods.

LOCAL AI & GENERATIVE WORKFLOWS

ComfyUI experimentation, model setup, image/video generation pipelines, prompt engineering, local inference configuration and GPU memory/resource troubleshooting.

MULTIMEDIA ENGINEERING & AUTOMATION

NexusVoid Gaming content production: OBS/NVIDIA capture, DaVinci Resolve and Adobe Premiere Pro editing, encoding, upscaling, HDR/SDR workflows and publishing automation concepts, including a multi-platform YouTube uploader.

AUDIO & DIGITAL SIGNAL PROCESSING

Practical parametric EQ and multiband tuning using Poweramp, Equalizer APO and FXSound; playback troubleshooting, DAC/IEM configuration, gaming audio and dialogue intelligibility.

LANGUAGES

English, Hindi and Telugu.